Recovering Latent Distance-Decay from Aggregate Mobility Data for Zero-shot Gravity Calibration

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Abstract

Accurate origin–destination (OD) flow estimation is fundamental to urban management, yet existing calibration approaches rely on increasingly restricted OD observations. This study investigates whether privacy-preserving aggregate mobility data preserve sufficient information to recover latent city-specific distance-decay parameters without relying on observed OD matrices. We derive aggregate trip-length distributions from sources such as Meta Movement Distribution Maps and formulate behavioral recovery as an exposure-corrected distribution matching problem in information space. Results across 50 US metropolitan areas show that recovered distance-decay parameters closely match locally calibrated references, yielding downstream flow predictions statistically equivalent to locally-calibrated models within a 0.01 Common Part of Commuters (CPC) margin. A Shapley analysis reveals that this recovered behavior accounts for 85.8% of the total predictive gain. By integrating these inferred parameters into a Projection–Inference Gravity Framework (PIGF), we enable zero-shot, survey-free OD estimation for data-scarce cities, demonstrating that aggregate mobility serves as a practical, privacy-preserving behavioral calibration signal.

Keywords: Urban mobility, distance decay calibration, gravity models, aggregate mobility data, zero-shot transfer, open-data modeling

1 Introduction

Accurate origin–destination (OD) flow prediction is fundamental to transportation planning, infrastructure development, and accessibility analysis (Barbosa et al. 2018; González et al. 2008; Song et al. 2010). Reliable OD information enables evidence-based infrastructure prioritization, travel demand forecasting, and urban mobility policy analysis across cities of all sizes. To model these spatial interactions, classical approaches such as gravity models (Wilson 1971; Tanner 1961) rely on a distance-decay parameter (β) to capture the behavioral friction of travel distance. More recent learning-based frameworks, including DeepGravity and graph neural networks (Simini et al. 2021; Hu et al. 2020), similarly depend on learning spatial friction representations from training data.

However, existing approaches share a common dependency: they require observed OD flows or individual mobility trajectories to estimate distance decay or to train their spatial interaction functions. Acquiring this localized mobility data remains a major practical challenge. Household travel surveys require substantial financial resources, limiting their coverage to a small fraction of cities worldwide (Egu and Bonnel 2020;

Hussain et al. 2021). Mobile-phone and GPS trajectory records, while richer in spatial and temporal detail, are frequently unavailable because of commercial licensing restrictions and increasingly strict privacy regulations. As a result, the majority of cities lack reliable local OD observations, rendering conventional OD-based calibration impossible.

In recent years, aggregate mobility products—such as Meta's Movement Distribution Maps (MDM) (Meta 2026)—have become increasingly available across platforms and regions. These products provide aggregated, population-level trip-length distributions (TLDs) rather than individual trajectories or OD matrices. Because they are publicly available, globally consistent, and designed to preserve privacy, aggregate mobility summaries offer a highly accessible data source in data-scarce and privacy-restricted environments (Pappalardo et al. 2023).

This availability creates a significant scientific opportunity, yet it leaves a fundamental knowledge gap: existing gravity-model calibration approaches rely on observed OD matrices, leaving it unclear whether aggregate mobility data preserve sufficient information to recover the city-specific distance-decay behavior associated with the underlying urban environment. In transportation science, travel behavior is recognized as an emergent response to urban infrastructure and spatial accessibility. However, whether privacy-preserving aggregate summaries retain sufficient information to recover expressive, two-parameter distance-decay functions (e.g., the Tanner deterrence function) remains uncharacterized.

To address this knowledge gap, this study asks the following central scientific question: *Does aggregate mobility data preserve sufficient information to recover city-specific distance-decay behavior?* From this question, we formulate our central hypothesis: *Aggregate trip-length distributions contain sufficient information to recover latent distance-decay parameters without requiring local OD observations. If successfully recovered, this latent behavioral parameter can replace conventional OD-based calibration, enabling transferable gravity models for cities without observed OD matrices.*

The rest of this paper is organized as follows. Section 2 reviews related work. Section 3 formalizes the problem setting. Section 4 presents the behavioral inference framework. Section 5 reports the experimental validation. Section 6 discusses the implications of our findings, and Section 7 addresses threats to validity.

2 Related Work

2.1 Mobility Modeling under Limited Behavioral Information

Accurate origin–destination (OD) prediction is a fundamental task in transportation planning, accessibility analysis, infrastructure design, and urban policy because it quantifies how travel demand is distributed between locations. Traditionally, reliable OD matrices are constructed from household travel surveys, census commuting records, mobile-phone trajectories, GPS traces, or smart-card transactions. While these data sources provide detailed mobility observations, they are often expensive to collect, infrequently updated, geographically incomplete, and increasingly constrained by privacy regulations (de Montjoye et al. 2013). Consequently, obtaining high-quality OD matrices remains one of the primary bottlenecks for applying mobility models, particularly in data-scarce cities and rapidly developing regions.

To alleviate this limitation, a wide range of OD prediction methods has been proposed. Classical spatial interaction models estimate mobility through analytical formulations (Wilson 1971; Simini et al. 2012), whereas more recent approaches employ deep learning to improve predictive performance by exploiting spatial representations learned from historical mobility data (Simini et al. 2021; Enaya et al. 2026). These advances have substantially improved OD prediction accuracy under data-rich conditions and expanded the ability of models to capture complex spatial interactions.

Despite their methodological differences, however, most existing approaches continue to depend on some form of mobility observation during model construction or calibration. Classical gravity models require observed OD flows to estimate city-specific distance-decay parameters (Lenormand et al. 2016), while supervised deep-learning models depend on historical OD matrices or trajectory datasets for training (Simini et al. 2021). Even recent transfer-learning and zero-shot approaches typically assume that representative

Table 1 Comparative analysis of human mobility models across structural and transferability dimensions.

Model Class	Decomposed Structure	Transferable Across Cities	Requires Target OD	Interpretable	Decay Explicitly Modeled
Gravity (Wilson 1971)	Yes	No	Yes	High	Yes
Radiation (Simini et al. 2012)	No	Yes	No	High	Yes
DeepGravity (Simini et al. 2021)	No	Limited	Yes	Low	No
Graph-based OD (MPGCN) (Hu et al. 2020)	No	Limited	Yes	Low	No
Graph Transfer Learning (GODDAG) (Rong et al. 2023a)	Partial	Yes	Yes	Low	No
Gravity-Informed Neural Models (Rong et al. 2023b)	Partial	Limited	Yes	Medium	Partial
PIGF (Ours)	Yes	Yes	No	High	Yes

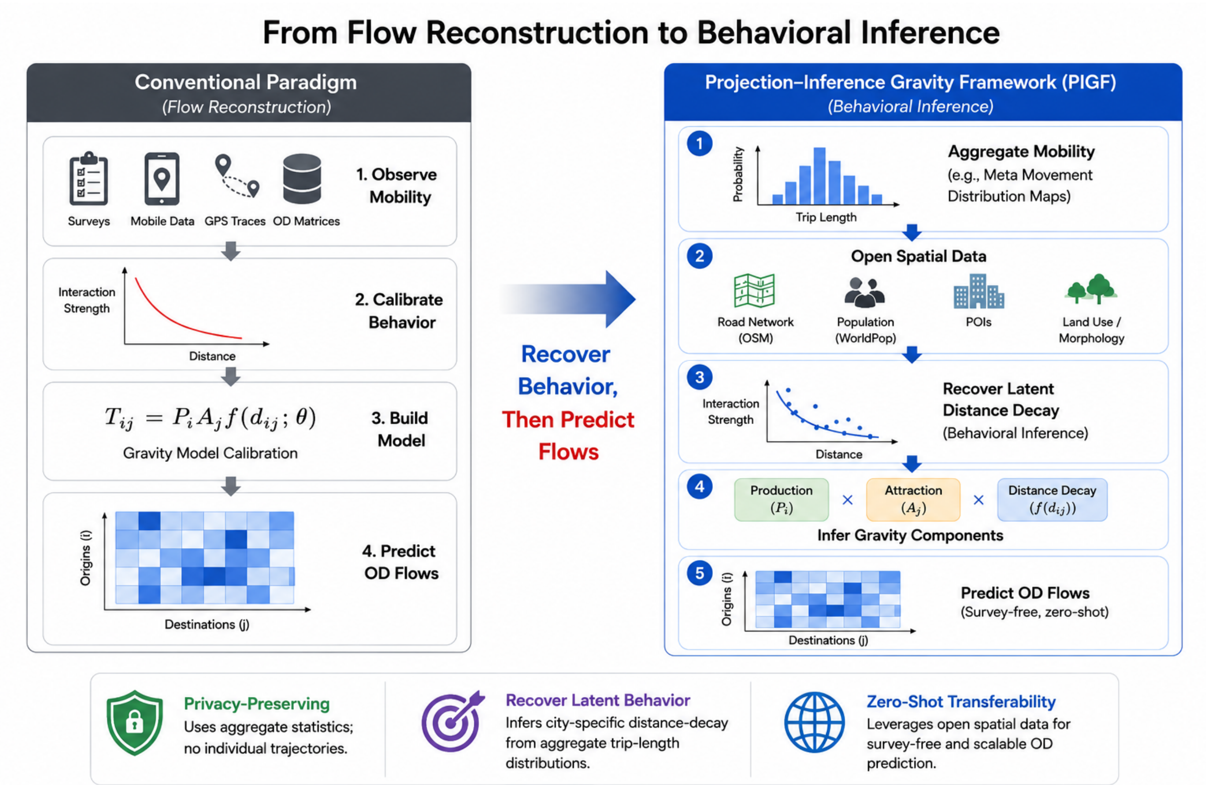

Fig. 1 Comparison between the conventional survey-based paradigm and the proposed Projection–Inference Gravity Framework (PIGF). Rather than reconstructing mobility flows from observed OD matrices, PIGF treats survey-free mobility modeling as a behavioral inference problem by recovering the latent city-specific distance-decay mechanism from aggregate mobility summaries before estimating OD flows.

mobility observations are available in source cities from which transferable knowledge can be learned (Yang et al. 2026). As a result, reliable mobility observations remain an essential prerequisite for most existing OD prediction frameworks.

These limitations motivate an important question: can meaningful mobility behavior be inferred without relying on conventional OD observations? Addressing this question requires understanding not only how mobility flows are predicted, but also how the underlying behavioral mechanism governing destination choice can be identified. This issue is closely related to the estimation of distance-decay functions, which forms the focus of the next section.

2.2 Distance-Decay as Behavioral Representation

Distance decay has long been recognized as the fundamental behavioral mechanism underlying spatial interaction. Within gravity-based mobility models, it describes how interaction probability decreases with increasing travel impedance and serves as the principal constraint governing destination choice. Since the pioneering work of Tanner and Wilson, distance-decay functions have formed the behavioral core of spatial interaction models and have been widely adopted in transportation planning, accessibility analysis, migration studies, and urban mobility forecasting (Tanner 1961; Wilson 1971; Fotheringham and O'Kelly 1989; Sen and Smith 1995). More recent studies continue to demonstrate that the choice of deterrence function substantially influences the accuracy and interpretability of mobility prediction models (Lenormand et al. 2016; Atwal et al. 2025).

To represent heterogeneous travel behavior, numerous functional forms have been proposed, including exponential, power-law, logistic, and Tanner-type deterrence functions. Among these, the Tanner function has become one of the most widely used formulations because it simultaneously captures short-distance attraction and long-distance attenuation using only two interpretable parameters. Recent research has further extended distance-decay calibration through statistical estimation, robust optimization, sparse regression, Bayesian inference, and physics-informed modeling to improve parameter stability and transferability while preserving behavioral interpretability (Flowerdew and Aitkin 1982; Lenormand et al. 2016; Rubio-Herrero and Muñuzuri 2023; Atwal et al. 2025).

Despite these methodological advances, existing calibration approaches share a common assumption: the availability of observed mobility flows. Classical gravity models estimate deterrence parameters directly from origin–destination matrices using maximum-likelihood estimation or least-squares optimization, while more recent data-driven approaches similarly rely on household travel surveys, mobile-phone trajectories, GPS traces, or complete OD observations to infer behavioral parameters (Wilson 1971; Flowerdew and Aitkin 1982; Rubio-Herrero and Muñuzuri 2023; Yang et al. 2026). Consequently, improvements have primarily focused on how to estimate distance-decay functions more accurately, rather than whether the underlying behavioral mechanism can be recovered without mobility observations.

This limitation becomes increasingly important as privacy-preserving mobility products replace individual trajectory datasets. Although aggregate mobility summaries no longer preserve origin–destination identities, they still retain the statistical distribution of travel distances, suggesting that they may encode the latent behavioral constraints governing destination choice. Whether such aggregate information is sufficient to recover city-specific distance-decay functions, however, remains largely unexplored. Addressing this question represents a critical prerequisite for enabling survey-free gravity calibration and forms the focus of the present study.

The possibility of recovering behavioral information from aggregate mobility summaries depends fundamentally on the information preserved in emerging aggregate mobility products. We therefore next review recent developments in privacy-preserving aggregate mobility datasets and their applications in urban mobility analysis.

2.3 Information Preservation in Aggregate Mobility

The growing availability of digital mobility data has substantially advanced the study of human movement, yet it has also intensified concerns regarding privacy protection, data governance, and the responsible sharing of individual travel records. Mobile-phone trajectories, GPS traces, and app-based location data contain detailed information about individual mobility behavior, making them highly valuable for transportation research while simultaneously raising significant privacy and ethical challenges. Consequently, recent years have witnessed a shift from publishing individual mobility records toward releasing privacy-preserving aggregate mobility products, which summarize collective travel behavior while preventing the reconstruction of individual trajectories (de Montjoye et al. 2013; Blondel et al. 2015).

Aggregate mobility products—including Meta's Movement Distribution Maps (MDM) (Meta 2026)—are becoming increasingly available across platforms and regions, providing privacy-preserving summaries of population travel behavior. Several large-scale aggregate mobility datasets have been developed following this paradigm. Examples include Meta MDM, which provide differentially private trip-length distributions aggregated over spatial regions; Google Community Mobility Reports, which summarize temporal changes in population activity across different place categories; and aggregated mobility datasets released through the Humanitarian Data Exchange (HDX) and related humanitarian initiatives for disaster response and population monitoring. Although these products differ in spatial resolution, aggregation strategy, and intended application, they share a common objective: preserving useful mobility information while substantially reducing privacy risks associated with individual trajectory data.

Because aggregate mobility products intentionally discard origin–destination identities, previous studies have primarily employed them for descriptive mobility analysis, accessibility assessment, epidemic modeling, disaster response, and large-scale urban monitoring (Oliver et al. 2020; Buckee et al. 2020; Tizzoni et al. 2022). Within this literature, aggregate mobility statistics are generally regarded as high-level indicators of collective movement patterns rather than as information suitable for calibrating behavioral mobility models. Consequently, their potential value for recovering the behavioral mechanisms underlying spatial interaction has received comparatively little attention.

Nevertheless, aggregate mobility products preserve an important characteristic that is directly relevant to gravity-based mobility modeling. Although they no longer identify individual origins or destinations, they retain the empirical distribution of travel distances across an urban system. Since distance-decay functions describe precisely how interaction probability varies with travel distance, these aggregate trip-length distributions may encode the latent behavioral information governing destination choice. Whether such aggregate statistics contain sufficient information to recover city-specific distance-decay functions, however, remains largely unexplored. Addressing this question is essential for enabling survey-free gravity calibration and motivates the behavioral inference framework proposed in this study.

While aggregate mobility products substantially reduce the dependence on individual trajectory data, existing mobility prediction approaches have only partially exploited the behavioral information preserved in these datasets. We therefore next review recent efforts toward survey-free and zero-shot mobility prediction and discuss their remaining limitations.

2.4 Behavior Transfer in Zero-Shot Prediction

The increasing scarcity of high-quality mobility observations has stimulated growing interest in survey-free and zero-shot mobility prediction. Rather than calibrating models independently for each city, these approaches aim to transfer mobility knowledge learned from data-rich regions to locations where conventional travel surveys are unavailable. By exploiting transferable spatial representations or behavioral regularities, they seek to reduce the dependence on locally collected OD observations while maintaining competitive predictive performance.

Recent studies have explored several transfer strategies. Deep learning approaches such as DeepGravity (Simini et al. 2021) demonstrate that complex nonlinear relationships between land use, population distribution, and travel demand can be learned directly from historical OD matrices, achieving substantial improvements over classical gravity models in data-rich environments. More recent graph-based and transfer-learning frameworks further improve cross-city prediction by learning transferable representations of urban mobility networks. For example, neuroGravity (Yang et al. 2026) reconstructs human mobility networks by combining neural representation learning with gravity-inspired inductive biases, while TransGM (Enaya et al. 2026) and related cross-city transfer models explicitly transfer mobility knowledge from source cities to previously unseen target regions.

These approaches significantly improve the practicality of mobility prediction under limited local data. However, they continue to rely on mobility observations during the learning process. Deep learning models require historical OD matrices or trajectory datasets for supervised training, whereas transfer-learning methods assume that representative mobility observations are available in one or more source cities from which transferable knowledge can be extracted. Consequently, although the target city may contain no OD observations, the predictive model itself remains fundamentally dependent on observed mobility behavior during model development.

An emerging line of research has begun incorporating physical principles into learning-based frameworks to improve interpretability and transferability. Physics-informed gravity models and hybrid neural architectures embed distance-decay constraints or spatial interaction mechanisms into deep learning models, reducing overfitting while preserving behavioral consistency (Atwal et al. 2025; Yang et al. 2026). Nevertheless, these methods still estimate behavioral mechanisms from observed mobility data rather than inferring them directly from privacy-preserving aggregate mobility summaries.

Therefore, despite substantial progress in survey-free and zero-shot mobility prediction, an important gap remains. Existing approaches primarily transfer behavioral knowledge learned from observed mobility data, whereas it remains unclear whether the behavioral mechanism itself can be recovered directly from aggregate mobility products without access to OD observations or trajectory records.

This remaining challenge motivates the central research gap addressed in this study. The following section synthesizes the limitations of existing literature and positions the proposed behavioral inference framework within the broader landscape of survey-free mobility prediction.

2.5 The Research Gap: From Learning to Inference

The preceding review demonstrates substantial progress in origin–destination prediction, spatial interaction modeling, aggregate mobility analysis, and survey-free mobility prediction. Classical gravity models have established distance decay as the fundamental behavioral mechanism governing spatial interaction, while recent deep learning and transfer-learning approaches have significantly improved prediction accuracy by exploiting increasingly rich mobility observations. At the same time, emerging privacy-preserving aggregate mobility products have expanded access to large-scale mobility information without exposing individual travel trajectories. Collectively, these developments have substantially advanced the state of urban mobility modeling. However, they also reveal a common conceptual limitation: behavioral information is consistently treated as something that must be learned from observed mobility flows rather than inferred directly from aggregate mobility statistics.

This distinction is fundamental. Existing gravity-based approaches estimate distance-decay functions from observed origin–destination matrices through model calibration (Wilson 1971; Flowerdew and Aitkin 1982; Sen and Smith 1995), whereas recent neural and transfer-learning methods learn transferable behavioral representations from historical mobility observations (Simini et al. 2021; Yang et al. 2026). Although these approaches differ considerably in modeling philosophy, they all assume that behavioral knowledge originates from some form of observed mobility data. Similarly, studies using privacy-preserving aggregate mobility products have primarily focused on descriptive mobility analysis, accessibility assessment, and population monitoring, rather than investigating whether these aggregate statistics themselves contain sufficient information for behavioral model calibration (Oliver et al. 2020; Buckee et al. 2020; Meta 2026).

These observations suggest that the central challenge in survey-free mobility prediction may not be the absence of OD matrices themselves, but rather the absence of a framework capable of recovering latent behavioral mechanisms directly from aggregate mobility information. If aggregate trip-length distributions preserve the statistical signature of distance-dependent travel behavior, then behavioral model calibration may be reformulated as an inverse inference problem instead of a conventional flow reconstruction problem. Such a perspective fundamentally differs from existing survey-free approaches, which primarily transfer behavioral knowledge learned elsewhere, by investigating whether the behavioral mechanism itself can be identified without first observing origin–destination flows.

Motivated by this perspective, this study formulates survey-free gravity calibration as a behavioral inference problem. Specifically, we investigate whether privacy-preserving aggregate trip-length distributions contain sufficient information to recover city-specific Tanner distance-decay functions and whether the recovered behavioral mechanism can support competitive OD prediction when integrated with transferable production estimation and analytical destination attraction.

By shifting the objective from flow reconstruction to behavioral inference, the proposed Projection–Inference Gravity Framework (PIGF) provides an interpretable and privacy-preserving alternative for mobility prediction in data-scarce environments.

The next section formalizes the problem setting and introduces the proposed behavioral inference framework.

3 Behavioral Inference Problem Formulation

This study formulates survey-free gravity calibration as the problem of recovering the latent city-specific distance-decay parameter from aggregate mobility observations. Once recovered, the parameter is integrated into a gravity model to estimate origin–destination flows. This formulation fundamentally shifts the focus from predicting an OD matrix directly to inferring the underlying spatial interaction behavior.

Consider a metropolitan area partitioned into a set of N spatial zones $\mathcal{V} = \{1, \ldots, N\}$. Let $\mathbf{T} \in \mathbb{R}_+^{N \times N}$ denote the unknown OD matrix, where each element T_{ij} represents the number of trips from origin zone i to destination zone j. Let $D = [d_{ij}]$ denote the inter-zonal distance matrix, where d_{ij} is the travel distance between zones i and j. For each zone i, an open spatial feature vector x_i is available, containing publicly accessible built-environment and demographic attributes derived from sources such as OpenStreetMap and WorldPop.

Unlike conventional OD prediction settings, we assume that no target-city OD matrix, household travel survey, GPS trajectory, mobile-phone record, or other individual mobility observation is available. Instead, the only mobility information provided for the target city is an aggregate Trip-Length Distribution (TLD), denoted by $P(d)$, which summarizes the population-level distribution of travel distances without revealing any origin–destination pairs or individual trajectories. Such aggregate mobility summaries are exemplified by products such as Meta Movement Distribution Maps (MDM).

Formally, our primary objective is to recover the latent distance-decay behavioral parameter θ:

$$\hat{\theta} = \arg\min_{\theta} \mathcal{L}\left(P(d), \hat{P}(d|\theta, D, X)\right) \tag{1}$$

where $\mathcal{L}$ is a loss function quantifying the divergence between the observed aggregate mobility $P(d)$ and the expected trip-length distribution given spatial features X and distance matrix D.

The secondary objective is to evaluate the validity of this recovered behavior by deploying it for survey-free OD estimation:

$$\hat{T} = f_{\text{gravity}}\left(D,\, X,\, \hat{\theta}\right) \tag{2}$$

The principal challenge of this formulation is that the observed TLD provides only a one-dimensional statistical summary of the underlying spatial interaction process. Recovering the latent distance-dependent interaction mechanism from such aggregate observations therefore constitutes an inverse inference problem. The following section details how our proposed framework solves this problem.

4 Behavioral Inference Framework

The methodological design of this study is grounded in a separable formulation of the classical spatial gravity framework. Rather than constructing a black-box model to map inputs to OD flows directly, we leverage the gravity model's structural decomposition to isolate the underlying behavioral mechanism from local, city-specific variables. This formulation allows us to test our central hypothesis: aggregate trip-length distributions contain sufficient information to recover latent behavioral parameters governing spatial interaction, enabling zero-shot gravity calibration without observed OD matrices.

4.1 Gravity Model Foundation

Spatial interaction has long been modeled using gravity-based formulations (Wilson 1971; Barbosa et al. 2018), which describe travel demand as the combined effect of trip generation, destination attractiveness, and travel impedance.

Under the singly constrained gravity formulation, the expected flow from origin zone i to destination zone j is expressed as

$$T_{ij} = O_i \frac{A_j\, f(d_{ij})}{\sum_k A_k\, f(d_{ik})}, \tag{3}$$

where O_i denotes the total trip production generated by origin zone i, A_j represents the attraction of destination zone j, d_{ij} is the travel distance between the two zones, and $f(d_{ij})$ is the distance-decay function. The normalization term ensures that the predicted outflows satisfy the production constraint, $\sum_j T_{ij} = O_i$.

A fundamental property of Eq. (3) is its **separable structure**. This decomposition isolates three distinct mechanisms of urban mobility:

- **Origin production** (O_i) captures the total travel demand generated by each zone, representing *demand generation*.
- **Destination attraction** (A_j) reflects the spatial distribution of opportunities, representing *opportunity distribution*.
- **Distance decay** $(f(d_{ij}))$ characterizes travelers' response to travel cost, representing the underlying *behavior*.

Under this paradigm, the distance-decay function is not merely a tunable mathematical parameter; it is a macroscopic signature of human behavior. While origin production and destination attraction are primarily determined by static, local attributes (e.g., population, employment, built environment), distance decay governs the dynamic spatial interaction.

This separation has an important implication for survey-free mobility modeling. By treating the gravity model as a generative process for spatial interaction, we can reformulate gravity calibration as the problem of recovering only the latent behavioral component (distance decay), while estimating production and attraction independently from publicly available spatial information. Consequently, if aggregate mobility observations preserve sufficient information about this behavior, the gravity model can be completely calibrated without requiring locally observed OD flows.

4.2 Aggregate Trip-Length Distributions as a Projection

The gravity formulation presented in the previous section provides a natural interpretation of aggregate trip-length distributions. Rather than representing an independent mobility descriptor, a trip-length distribution can be viewed as the distance-domain projection of the underlying origin–destination interaction process. This interpretation establishes the theoretical connection between aggregate mobility observations (Meta 2026; Pappalardo et al. 2023) and the latent distance-decay mechanism that governs spatial interaction.

Let the travel distance domain be partitioned into a set of distance intervals $\{b\}_{b=1}^{K}$. The aggregate trip-length distribution is defined as

$$P(d_b) = \sum_{i,j} T_{ij}\, \mathbf{1}(d_{ij} \in b), \tag{4}$$

where $P(d_b)$ denotes the proportion of trips whose travel distance falls within interval b, and $\mathbf{1}(\cdot)$ is the indicator function identifying origin–destination pairs associated with that interval.

Unlike the OD matrix, which explicitly preserves the identities of origins and destinations, the projection removes all location-specific information and retains only the statistical distribution of travel distances. Consequently, aggregate trip-length distributions cannot directly reconstruct the complete OD matrix. Nevertheless, because the gravity formulation separates production, attraction, and distance-dependent behavioral effects, the projection continues to preserve the characteristic statistical signature of the distance-decay component.

However, the observed trip-length distribution is not equivalent to the distance-decay function itself. The projection is simultaneously influenced by urban geometry, the spatial distribution of opportunities, and the unequal number of OD pairs contributing to each distance interval. These structural effects introduce systematic **exposure biases** (Lenormand et al. 2016), implying that identical behavioral distance-decay functions may produce different aggregate trip-length distributions across cities with different spatial configurations. Therefore, directly fitting a deterrence function to the observed histogram would generally yield biased parameter estimates.

This observation transforms survey-free gravity calibration into an **inverse inference problem**. Rather than estimating the complete OD matrix, the objective becomes identifying the latent distance-dependent behavioral mechanism whose exposure-corrected projection best explains the observed aggregate trip-length distribution.

4.3 Distance-Decay Parameterization

To represent travelers' sensitivity to travel distance, we adopt the **Tanner deterrence function** (Tanner 1961), a flexible two-parameter formulation that combines power-law and exponential decay and has been shown to outperform single-family alternatives across diverse urban settings (Verma and Ukkusuri 2025). The distance-decay function is expressed as

$$f(d_{ij};\,\theta) = (d_{ij} + \delta)^{-\alpha} \exp(-\beta d_{ij}), \qquad (5)$$

where $\theta = (\alpha,\, \beta)$ denotes the unknown behavioral parameters. The power-law exponent $\alpha \geq 0$ controls the concentration of short-to-medium distance trips, whereas the exponential parameter $\beta > 0$ governs the attenuation of long-distance interactions.

Importantly, from a transportation science perspective, the decay parameters $\theta = (\alpha, \beta)$ do not represent intrinsic or fixed personal traits of individual residents. Rather, they parameterize an emergent property of **human–environment interaction**: transit network layout and urban land use determine spatial accessibility, which conditions travelers' destination choices, ultimately manifesting as the observed empirical distance-decay response (Transit/Infrastructure $\rightarrow$ Spatial Accessibility $\rightarrow$ Travel Behavior $\rightarrow$ (α, β)).

To avoid numerical singularities when travel distance approaches zero while maintaining consistent behavior across different spatial partitioning schemes, we introduce an **adaptive geometric offset**

$$\delta = 0.5\,\bar{R}_{\text{zone}}, \qquad \bar{R}_{\text{zone}} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\frac{\text{Area}_i}{\pi}} \qquad (6)$$

where $\bar{R}_{\text{zone}}$ is the average equivalent radius of all spatial zones. Unlike a fixed numerical constant, the adaptive offset automatically scales with the characteristic spatial resolution of the study area, improving numerical stability while preserving the physical interpretation of the recovered distance-decay parameters. During implementation, travel distances are additionally lower-bounded by a small positive constant (10^{-6}) to prevent floating-point instability without affecting practical inter-zonal distances.

4.4 Exposure Correction & Information-Theoretic Distribution Matching

Building upon the projection formulation above, recovering the distance-decay behavior is formulated as an information-theoretic distribution matching problem via minimum relative entropy. Rather than treating parameter fitting as a data-intensive flow-fitting exercise, the proposed framework explicitly separates structural spatial exposure from latent traveler preferences.

For each distance interval b, let $E(b)$ denote the corresponding **spatial exposure**, defined as the total structural interaction opportunity associated with all origin–destination pairs whose travel distances fall within interval b. Exposure reflects purely physical and geometric characteristics of the metropolitan region—including zonal layout and opportunity distributions—and is completely independent of travelers' latent distance sensitivity.

Under the gravity formulation, the predicted probability $\pi_b(\theta)$ that a random travel interaction occurs within distance bin b is given by

$$\pi_b(\theta) = \frac{E(b)\, f(d_b;\, \theta)}{\sum_{b'} E(b')\, f(d_{b'};\, \theta)}, \qquad (7)$$

which naturally isolates the behavioral deterrence function $f(d_b;\, \theta)$ from geometric exposure $E(b)$.

Given an empirical aggregate trip-length distribution (TLD) represented as normalized travel fractions $b = (b_1, b_2, \ldots, b_K)^\top$ where $\sum_{k=1}^{K} b_k = 1.0$ (such as telemetry provided by Meta Movement Distribution Maps), behavioral parameter recovery is achieved by minimizing the Kullback–Leibler (KL) divergence (relative entropy) between the empirical distribution b and the model distribution π_θ:

$$D_{\mathrm{KL}}(b \parallel \pi_\theta) = \sum_{k=1}^{K} b_k \log\left(\frac{b_k}{\pi_k(\theta)}\right) = -H(b)+H(b,\pi_\theta), \quad (8)$$

where $H(b) = -\sum b_k \log b_k$ is the empirical entropy (constant with respect to θ), and $H(b,\pi_\theta)$ represents the cross-entropy:

$$H(b,\pi_\theta) = -\sum_{k=1}^{K} b_k \log \pi_k(\theta). \quad (9)$$

Minimizing relative entropy $D_{\mathrm{KL}}(b \parallel \pi_\theta)$ is therefore mathematically equivalent to minimizing the cross-entropy objective:

$$\hat{\theta} = \arg\min_{\theta}\ H(b,\pi_\theta) = \arg\min_{\theta}\left[-\sum_{k=1}^{K} b_k \log \pi_k(\theta)\right]. \quad (10)$$

When aggregate trip counts y_b with total sample size $N = \sum y_b$ are available, minimizing cross-entropy $H(b,\pi_\theta)$ is mathematically equivalent to maximizing multinomial log-likelihood up to a positive constant scale factor N ($\log \mathcal{L}(\theta) = -N \cdot H(b,\pi_\theta)$). Operating directly on normalized fractions b_k provides parameter recovery that is strictly scale-invariant and fully compatible with privacy-preserving telemetry products where total trip counts N are intentionally omitted.

To guarantee the positivity constraints $\alpha \geq 0$, $\beta > 0$, optimization is performed in the unconstrained logarithmic parameter space using the L-BFGS-B algorithm with multiple randomized initializations. The first run is initialized from a moment-matched baseline, while remaining restarts introduce Gaussian perturbations to avoid local minima.

Unlike conventional gravity calibration requiring disaggregate OD flow matrices, the proposed formulation isolates and infers only the latent behavioral mechanism. By explicitly correcting for spatial exposure, gravity calibration is transformed from a data-intensive flow-fitting problem into an information-theoretically grounded distribution matching problem requiring only aggregate trip-length fractions. The overall architecture is illustrated in Fig. 2.

4.5 Statistical Foundations of Behavioral Recovery

To establish the statistical foundations of the proposed behavioral recovery module, we clarify the relationship between classical gravity model calibration and the proposed aggregate distribution matching objective through three progressive stages.

4.5.1 Classical Poisson Gravity Calibration

In classical spatial interaction literature, origin–destination travel flows T_{ij} are modeled as discrete count observations. Consequently, Poisson likelihood (and related count-based likelihood formulations) serves as the classical statistical foundation for gravity calibration from observed OD trip counts (Flowerdew and Aitkin 1982; Sen and Smith 1995; Lenormand et al. 2016). Under a Poisson specification, deterrence parameters are estimated by maximizing the Poisson log-likelihood over individual OD matrix cells:

$$\log \mathcal{L}_{\mathrm{Poisson}}(\theta; \mathbf{T}) = \sum_{i,j}\left[T_{ij}\log \hat{T}_{ij}(\theta) - \hat{T}_{ij}(\theta) - \log(T_{ij}!)\right], \quad (11)$$

where $\hat{T}_{ij}(\theta)$ represents the gravity model prediction. While mathematically rigorous for complete flow matrices, Poisson likelihood estimation inherently depends on observing full, unaggregated $N \times N$ OD matrices (T_{ij}), which are unavailable under privacy-preserving telemetry constraints.

4.5.2 Aggregate Histogram Likelihood

In contrast, our framework operates on aggregate trip-length distributions rather than individual trip counts. When individual mobility traces are unobservable due to privacy restrictions, telemetry products (such as Meta Movement Distribution Maps) release normalized travel fractions $\mathbf{b} = (b_1, b_2, \ldots, b_K)^\top$, where $b_k = y_k/N$, $\sum_{k=1}^{K} b_k = 1.0$, and y_k represents the unobserved trip count in distance bin k.

Because individual OD counts are unobservable, the calibration target shifts from matrix-level Poisson likelihood to bin-level distribution matching. If histogram count vector $\mathbf{y}$ were observed, the natural log-likelihood under a multinomial sampling assumption over distance bins would be

$$\log \mathcal{L}_{\mathrm{Multi}}(\theta; \mathbf{y}) = \log \left(\frac{N!}{\prod_{k=1}^{K} y_k!} \right) + \sum_{k=1}^{K} y_k \log \pi_k(\theta) = C + N \sum_{k=1}^{K} b_k \log \pi_k(\theta), \tag{12}$$

where C is a constant independent of parameter vector $\theta = (\alpha, \beta)$.

4.5.3 Cross-Entropy Equivalence & Normalization Invariance

Rearranging Eq. (12) connects the multinomial likelihood directly to the proposed cross-entropy objective:

$$\log \mathcal{L}_{\mathrm{Multi}}(\theta; \mathbf{y}) = C - N \cdot H(\mathbf{b}, \boldsymbol{\pi}_\theta) = C - N \left[H(\mathbf{b}) + D_{\mathrm{KL}}(\mathbf{b} \parallel \boldsymbol{\pi}_\theta) \right], \tag{13}$$

where $H(\mathbf{b}, \boldsymbol{\pi}_\theta) = -\sum b_k \log \pi_k(\theta)$ is the cross-entropy, $H(\mathbf{b})$ is the empirical entropy of the aggregate telemetry, and $D_{\mathrm{KL}}(\mathbf{b} \parallel \boldsymbol{\pi}_\theta)$ is the Kullback–Leibler (KL) divergence.

Taking the gradient with respect to parameter vector θ, the score equation satisfies

$$\nabla_\theta \log \mathcal{L}_{\mathrm{Multi}}(\theta; \mathbf{y}) = -N \cdot \nabla_\theta H(\mathbf{b}, \boldsymbol{\pi}_\theta) = 0 \iff \nabla_\theta H(\mathbf{b}, \boldsymbol{\pi}_{\hat{\theta}}) = 0 \quad (\text{for } N > 0). \tag{14}$$

Poisson likelihood serves as the classical statistical foundation for gravity calibration from observed OD trip counts. In contrast, our framework operates on aggregate trip-length distributions rather than individual trip counts. Consequently, the proposed objective is formulated as cross-entropy minimization over normalized distance distributions, which is mathematically equivalent to multinomial maximum likelihood estimation up to a positive constant scaling factor (N). This property—**Normalization Invariance**—ensures that moving from raw OD counts $\mathbf{T} \rightarrow$ aggregate histogram counts $\mathbf{y} \rightarrow$ normalized histogram fractions $\mathbf{b}$ preserves the exact optimal behavioral parameters (α, β). Thus, the proposed framework can operate directly on privacy-preserving aggregate telemetry without requiring disaggregate OD matrices or raw trip counts.

4.6 Transferable Origin Production Estimation (O_i)

Within the gravity formulation, origin production represents the total number of trips generated by each spatial zone and is conceptually independent of the behavioral interaction mechanism recovered in the previous section. While distance decay governs travelers' destination choices, origin production is primarily determined by relatively stable demographic and built-environment characteristics. This distinction enables the two components to be estimated independently.

For each spatial zone i, we construct an open spatial feature vector $\mathbf{x}_i$ consisting of publicly available variables that describe local urban characteristics. These features include population, land-use composition, point-of-interest (POI) distributions, road-network density, and other built-environment indicators derived from open geospatial datasets such as OpenStreetMap and WorldPop (Haklay and Weber 2008; Tatem 2017). Because these variables are available for virtually all cities without requiring mobility observations, they provide a transferable basis for estimating trip generation.

The mapping between spatial features and origin production is learned from source cities using a Gradient Boosting Decision Tree (GBDT) regression model (Friedman 2001),

$$\hat{O}_i = g(\mathbf{x}_i), \tag{15}$$

where $g(\cdot)$ denotes the learned nonlinear regression function and $\hat{O}_i$ is the predicted trip production for zone i. GBDT is adopted because of its strong predictive performance on heterogeneous tabular data, its ability to model nonlinear interactions without extensive feature engineering, and its robustness under cross-city prediction settings. We train the GBDT on a parsimonious set of **6 core macroscopic features**: total population (P_i), total POI count (POI_i), zone area (Area_i), population density ($\rho_{pop,i}$), POI density ($\rho_{poi,i}$), and road density (rd_i).

These six features were selected from an initial 37-candidate set using a cross-city SHAP stability analysis, showing no performance degradation compared to the full feature set. This parsimonious selection is grounded in the dimensional scaling of urban physics and Occam's razor, thereby preventing the model from overfitting to source-domain geometries. Hyperparameters and details of the SHAP-based feature selection are provided in Appendix C.

Once trained on source-city data, the regression model is directly applied to the target city

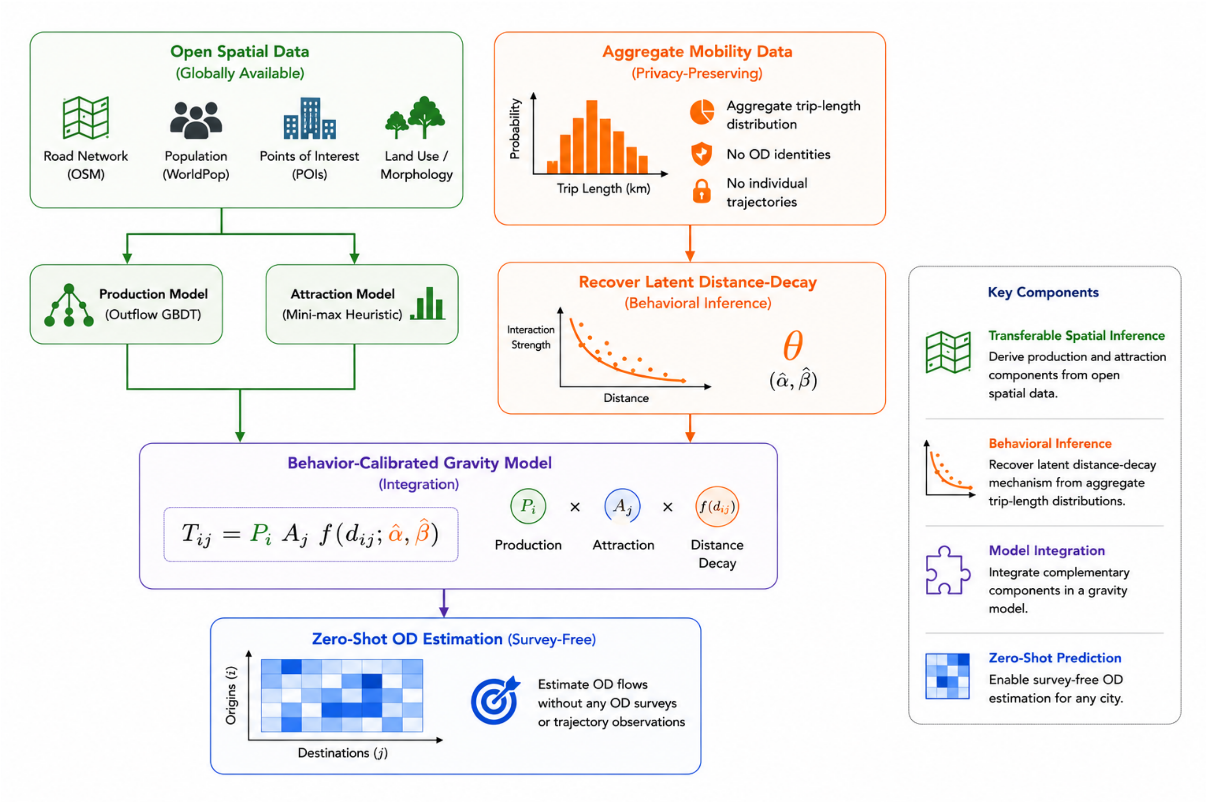

Fig. 2 Technical realization of the behavioral inference paradigm. Instead of calibrating gravity models from observed origin–destination (OD) flows, PIGF recovers the latent city-specific distance-decay mechanism from aggregate trip-length distributions and combines it with transferable production and attraction components derived from open spatial data to perform survey-free, zero-shot OD estimation.

without further adaptation or fine-tuning. Consequently, origin production can be estimated under strict zero-shot conditions using only publicly available spatial information, eliminating the need for local travel surveys or historical OD observations.

4.7 Destination Attraction Estimation (A_j)

Unlike origin production, destination attraction reflects the relative ability of each spatial zone to attract trips from all origins within the urban system. Its value is therefore determined not only by the intrinsic characteristics of an individual zone but also by the overall spatial distribution of opportunities across the entire city. This dependence makes destination attraction considerably less transferable than origin production and limits the applicability of conventional supervised learning approaches.

Instead of learning attraction from historical destination inflows, the proposed framework estimates attraction directly from publicly available urban opportunity indicators. Let $\mathbf{x}_j$ denote the feature vector associated with destination zone j. The attraction score is estimated through a training-free, scale-invariant attraction heuristic:

$$\hat{A}_j = \frac{1}{2}\left[\mathrm{minmax}(P_j) + \mathrm{minmax}(\mathrm{POI}_j)\right] \quad (16)$$

where $\mathrm{minmax}(x_j) = (x_j - \min_k x_k)/(\max_k x_k - \min_k x_k + 10^{-9})$, P_j is the gridded population count, and POI_j is the OpenStreetMap POI count. This formulation normalizes absolute scale differences across cities while preserving local opportunity hierarchies, ensuring that each indicator contributes proportionally during attraction estimation.

This analytical estimation strategy offers three practical advantages. First, it is entirely survey-free, requiring only publicly available urban information. Second, it is training-free, avoiding dependence on historical destination-flow observations. Third, because the attraction scores are computed from normalized indicators, the resulting formulation remains scale-invariant, facilitating direct application across cities with substantially different spatial characteristics. In Section 5.2, the ablation analysis validates that this training-free heuristic contributes significantly to the downstream prediction accuracy.

4.8 Survey-Free Origin–Destination Flow Generation

The preceding sections established the methods to infer the three necessary components of the gravity formulation: the latent distance-decay behavior, transferable origin production, and analytical destination attraction. Once these components are recovered, they can be integrated within the gravity framework. At this stage, generating the complete origin–destination flow matrix is a downstream consequence of the behavioral recovery, rather than the primary fitting objective.

Substituting the recovered behavioral parameters and the estimated production and attraction terms into Eq. (3) yields

$$\hat{T}_{ij} = \hat{O}_i \frac{\hat{A}_j \, f(d_{ij}; \hat{\theta})}{\sum_k \hat{A}_k \, f(d_{ik}; \hat{\theta})}, \tag{17}$$

where $\hat{O}_i$ denotes the estimated origin production, $\hat{A}_j$ represents the estimated destination attraction, and $\hat{\theta} = (\hat{\alpha}, \hat{\beta})$ are the recovered Tanner parameters.

Equation (17) preserves the production constraint $\sum_j \hat{T}_{ij} = \hat{O}_i$ while allocating trips among competing destinations according to both their relative opportunities and the recovered behavioral response to travel distance.

Unlike conventional gravity calibration, which directly fits observed OD matrices to minimize flow error, PIGF infers the interaction mechanism from the information source most directly associated with its behavioral meaning. Because no target-city OD matrices, household travel surveys, or trajectory records are required, Eq. (17) enables survey-free and zero-shot OD generation using only publicly accessible aggregate information.

4.9 Properties of the Behavioral Inference Framework

The proposed PIGF possesses several methodological characteristics that distinguish it from existing gravity calibration and supervised OD prediction approaches.

Behavioral Identifiability. The core novelty of the proposed framework is its ability to recover latent, city-specific distance-decay parameters directly from one-dimensional aggregate trip-length distributions by explicitly correcting for spatial exposure, without requiring full OD matrices.

Survey-Free Calibration. Unlike conventional gravity models that require locally observed OD matrices for parameter calibration, PIGF estimates all gravity components without using any target-city mobility observations.

Component-Wise Inference. Instead of learning the complete OD matrix as a black-box mapping, PIGF decomposes spatial interaction into three interpretable components—origin production, destination attraction, and distance decay—and estimates each component using the information source most directly associated with its physical or behavioral meaning.

Transferability. All components of PIGF rely exclusively on publicly available or aggregate information, allowing the framework to be directly deployed in previously unseen cities without recalibration using local mobility observations. This property naturally enables zero-shot OD prediction in data-scarce environments.

4.10 Computational Complexity

The computational complexity of the proposed framework is dominated by the final gravity-based OD generation step. Distance-decay recovery operates over aggregated distance bins and therefore scales linearly with the number of distance intervals. Origin production prediction and destination attraction estimation are both linear in the number of spatial zones after model training. Consequently, the overall computational cost is dominated by evaluating pairwise gravity interactions, resulting in a complexity of

Algorithm 1: Projection–Inference Gravity Framework (PIGF)

Input: Source cities data $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}_{c=1}^{C}$; Target city spatial features $\mathbf{X}^{(t)}$; Target city aggregate trip-length histogram $\{y_b\}_{b=1}^{K}$ (or national prior if unavailable)
Output: Predicted Target City OD Matrix $\hat{\mathbf{T}}^{(t)}$

Phase 1: Offline Training (Source Cities)
1. **Production Model Training**: Train the outflow model $g(\mathbf{x})$ (GBDT) on pooled source-city spatial features $\mathbf{X}^{(c)}$ and ground-truth outflows $O_i^{(c)}$.

Phase 2: Online Behavioral Inference & Zero-Shot Transfer (Target City)
2. **Latent Behavioral Inference**: Recover target-city decay parameters $(\hat{\alpha}, \hat{\beta})$ via cross-entropy minimization on the target city's aggregate TLD fractions $\{b_k\}_{b=1}^{K}$. (If unavailable, fall back to the source-derived national prior).
3. **Production Estimation**: Predict target city outflows: $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$.
4. **Attraction Mapping**: Compute target city zone attractions using the training-free heuristic: $\hat{A}_j^{(t)} = \frac{1}{2}\left[\mathrm{minmax}(P_j^{(t)}) + \mathrm{minmax}(\mathrm{POI}_j^{(t)})\right]$.
5. **Survey-Free Target Prediction**:
   a. Compute adaptive target offset: $\delta^{(t)} = 0.5 \times \bar{R}_{\mathrm{zone}}^{(t)}$.
   b. Fuse components and normalize outflows using Eq. (17) with recovered parameters $(\hat{\alpha}, \hat{\beta})$.

$$O(N^2),$$

where N denotes the number of spatial zones. This complexity is consistent with conventional gravity models while eliminating the computationally intensive calibration process based on observed OD matrices.

4.11 Closing Remarks

The proposed Projection–Inference Gravity Framework reformulates survey-free OD prediction as a component-wise gravity inference problem rather than a direct origin–destination prediction task. Instead of learning the entire OD matrix from historical mobility observations, PIGF independently recovers the behavioral distance-decay mechanism from aggregate trip-length distributions, estimates origin production from transferable open spatial features, and derives destination attraction analytically from publicly available urban opportunity indicators. By integrating these independently inferred components within a unified gravity formulation, the framework enables interpretable and survey-free urban OD prediction under strict zero-shot conditions.

If the proposed theoretical formulation is valid, three empirical consequences should follow. First, aggregate trip-length distributions should enable accurate recovery of city-specific distance-decay parameters. Second, the recovered behavioral component should explain a substantial proportion of OD prediction performance relative to the remaining gravity components. Third, integrating the three inferred components should produce competitive survey-free OD predictions across diverse urban environments. The following section evaluates these hypotheses through comprehensive experiments.

4.12 Data Sources

We utilize three globally available open data sources to compile features for our models: built-environment spatial descriptors (OpenStreetMap POIs and road density), gridded demography (WorldPop), and aggregate movement telemetry maps (Meta Movement Distribution Maps (Meta 2026)). Ground-truth origin-destination (OD) flows between census tracts are obtained from the LODES commuting dataset covering 50 US metropolitan areas (25 source cities for GBDT production training and 25 held-out target cities for testing). The target OD matrices are kept completely blind to the model components and are used exclusively as ground truth for validation.

4.13 Behavioral Recovery from Real Aggregate Mobility Products

Under the proposed theoretical framework, any aggregate data source representing the probability distribution of human movement distances is sufficient to recover city-specific distance-decay parameters. In this paper, we use the term Trip-Length Distribution (TLD) to denote aggregate travel distance distributions, including products such as the HDX Movement Distribution dataset. To evaluate this formulation under real-world telemetry constraints, we utilize Meta's Movement Distribution Maps (MDM) as a representative evidence of such platform-derived aggregate mobility indicators.

Meta MDM provides aggregate travel fractions (representing empirical TLDs) at the county level grouped into 3 physical distance bins: $[0, 10 \text{ km}]$, $(10, 100 \text{ km}]$, and > 100 km. To harmonize these county-level travel fractions into tract geometries for calibration:

1. County-level trip-length distributions are averaged using WorldPop-derived population shares as weights: $p_{\text{metro}}(b) = \sum_{c \in \mathcal{C}} w_c \cdot p_c(b)$, where w_c is the population share of intersecting county c.
2. Tract pairs (i, j) are mapped to these 3 bins based on their centroid distances d_{ij}.
3. The predicted probability of a flow falling into Meta bin b is computed by summing and normalizing predicted tract-level flows in that bin: $P_\theta(b) = \sum_{i,j:d_{ij} \in \text{Range}(b)} \hat{T}_{ij}(\theta) / \sum_{i,j} \hat{T}_{ij}(\theta)$. This normalization bridges the continuous gravity formulation with Meta's 3-bin telemetry.

4.14 Exploratory Urban Morphology Categorization

To investigate how physical geography modulates the role of distance decay in spatial interactions, the 25 target cities are classified into four morphological groups based on their urban structure and geography (González et al. 2008). This classification is summarized in Table 2.

5 Experimental Validation

5.1 RQ1: Can latent behavior be recovered?

We begin by evaluating the fundamental hypothesis: can aggregate mobility preserve the latent distance-decay behavior? We assess this by analyzing the identifiability of behavioral parameters across 50 US metropolitan areas. The recovery metrics reveal strong structural preservation: the power-law exponent α (capturing short-distance deterrence) exhibits high consistency across cities ($r = 0.909$), while the exponential rate β (capturing long-distance decay) shows a tight absolute margin of error (MAE $= 0.0026$) (Table 3). These recovered parameters indicate that aggregate mobility data preserve the dominant behavioral signature governing city-specific distance-dependent interactions.

To validate whether this recovered behavior is meaningful, we deploy it for downstream spatial interaction modeling under oracle outflow control. Employing the distance-decay behavior inferred solely from aggregate TLDs yields OD prediction accuracy nearly identical to that obtained using parameters calibrated directly on full zone-to-zone OD surveys (Table 4). Specifically, the average difference in the Common Part of Commuters (CPC) index between the survey-free recovered behavior and survey-calibrated behavior is a negligible 0.11%. This demonstrates that the recovered behavioral parameters are not just mathematically identifiable, but physically valid for reproducing spatial interactions.

5.1.1 Validation via Trip-Length Distributions (Meta MDM Case Study)

To evaluate the practical feasibility of recovering distance decay from real-world TLDs, we apply our exposure-corrected distribution matching framework to Meta's Movement Distribution Maps prior (`Meta MDM Prior (3-Bin)`) as an empirical case study across the 25 target cities. We compare its downstream CPC performance under oracle outflows against models calibrated on LODES commuting data aggregated into 3 bins (`LODES Baseline (3-Bin)`) and 20 equal-width bins (`LODES Oracle (20-Bin)`).

Table 2 Morphological classification of target cities.

Morphology Group	Cities	Description
Dense / Transit ($n = 5$)	Boston, Long Beach, New York, Philadelphia, Washington DC	High-density urban cores; extensive transit networks; weaker distance decay influence.
Sprawling / Grid ($n = 10$)	Atlanta, Dallas, El Paso, Houston, Jacksonville, Las Vegas, Mesa, Omaha, San Antonio, Tulsa	Low-to-medium density; regular road grids; car-dependent mobility with uniform decay.
Polycentric ($n = 7$)	Albuquerque, Charlotte, Detroit, Louisville, Milwaukee, Raleigh, San Jose	Multiple employment centers; flows cluster around local nodes rather than a single center.
Coastal ($n = 3$)	Portland, Seattle, Virginia Beach	Coastline/topographic barriers constrain the road network, amplifying distance friction.

Table 3 Decay parameter recovery performance metrics across the 25 target cities.

Parameter	Pearson r	MAE	Median Relative Error (%)	Cities within ±10%	Practic tion
Power-law exponent (α)	0.909	0.035	1.7%	100.0%	Strong s across c
Exponential decay rate (β)	0.427	0.0026	0.0%	72.0%	Small despite correlat

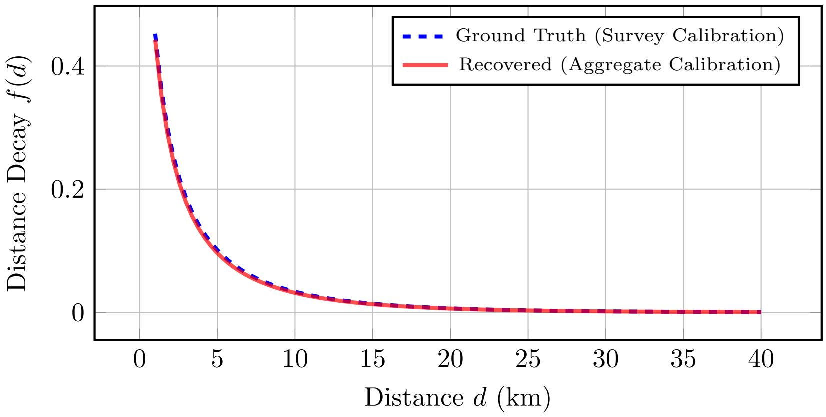

Fig. 3 Conceptual comparison of the Tanner distance-decay function $f(d) = (d+\delta)^{-\alpha} \exp(-\beta d)$. The parameters recovered from aggregate trip-length distributions (solid red) closely approximate the parameters calibrated on complete ground-truth survey data (dashed blue). The recovered function effectively preserves both the power-law concentration at short distances and the exponential attenuation at long distances, demonstrating robust behavioral recovery.

As shown in Table 4, the survey-free `Meta MDM Prior (3-Bin)` calibration yields a mean CPC of 0.7080 across the 25 target cities, compared to 0.7348 for the 3-bin survey baseline and 0.7403 for the 20-bin survey oracle.

Histogram compression is not the primary source of performance loss. When the original LODES commuting data is compressed from 20 bins to the same three physical distance bins used by Meta MDM (0–10 km, 10–100 km, > 100 km), CPC decreases only from 0.7403 to 0.7348 (−0.0055). This indicates that the coarse 3-bin representation preserves most of the information required for distance-decay recovery. In contrast, replacing the survey-derived histogram with real Meta MDM telemetry further reduces CPC from 0.7348 to 0.7080 (-0.0268). Therefore, most of the observed degradation is attributable to source mismatch and telemetry aggregation effects rather than to the limited number of bins. These results support the central claim that aggregate trip-length distributions remain highly informative for distance-decay calibration even under extremely compressed representations.

To evaluate performance under extreme data scarcity where target trip-length distributions (TLDs) are completely missing, we analyze the National Decay Fallback strategy. Under this strategy, the target city's distance-decay parameters are set to a national prior defined as the median parameter estimates fitted across the set of source cities. Under oracle outflows, this national prior model achieves a mean CPC of 0.7398, and under GBDT-predicted outflows, it yields a mean CPC of 0.6896 (as reported in Table 6).

Furthermore, we observe that the localized Meta MDM calibration is sensitive to specific local outlier dynamics. For instance, the metropolitan area of Atlanta serves as a prominent outlier in our evaluation, exhibiting a localized performance drop compared to the 3-bin LODES baseline. A detailed discussion of this outlier behavior is provided in Appendix A.2.

5.1.2 Decay Bin Sensitivity Analysis

To investigate how the granularity of distance binning affects model performance, we sweep the number of distance bins K across $\{3, 5, 10, 20\}$. The results show that predictive performance improves substantially from $K = 3$ to $K = 5$ and then exhibits diminishing returns, with mean CPC values of 0.7174 ($K = 3$), 0.7346 ($K = 5$), 0.7394 ($K = 10$), and 0.7403 ($K = 20$). Detailed comparisons and sensitivity analysis are provided in Appendix A.5.

5.2 RQ2: Is recovered behavior informative?

Having established that distance-decay behavior can be recovered from aggregate mobility data, we now investigate its relative importance within the spatial interaction process. We execute a game-theoretic Shapley value analysis and a factorial ablation study to quantify how much of the predictive information originates from the behavioral component versus structural characteristics (production and attraction).

5.2.1 Factorial Ablation and Shapley Value Analysis

To evaluate the relative importance of each gravity mechanism in zero-shot transfer scenarios, we perform a factorial ablation study across the 25 target cities. Each component is disabled in turn and replaced by a uniform-weight fallback, isolating its marginal contribution to the gravity model formulation.

Based on this ablation, we execute a game-theoretic analysis using Shapley values (Shapley 1953) computed across all $2^3 = 8$ possible component coalitions. Relative to a uniform-distribution baseline (assigning equal flow weight to all zone pairs; CPC 0.4263), the Shapley values attribute a mean CPC gain of 0.2258 (accounting for 85.8% of the total gain) to the distance-decay behavior $f(d)$, a gain of 0.0208 (7.9% of the total gain) to the structural attraction heuristic A_j, and a gain of 0.0167 (6.4% of the total gain) to the origin production component O_i. The baseline model for the factorial ablation presented in Table 5 is the complete gravity model (mean CPC of 0.6896). Removing the distance-decay behavior causes the performance to plummet to a CPC of 0.4489, representing a nearly complete loss of predictive power.

5.2.2 Interpretation of Behavioral Dominance

These results represent a core scientific discovery regarding the mechanics of urban mobility. The 85.8% Shapley attribution demonstrates that the most predictive information for modeling spatial interaction originates from behavioral distance friction, drastically outweighing the structural variables governing local production and attraction. In other words, knowing *how* people react to travel distance is vastly more important than precisely estimating *where* opportunities are located. This finding validates our primary focus on recovering latent distance-decay parameters: because

Table 4 Downstream CPC performance on the 25 held-out target cities under different decay calibration strategies (Oracle Outflows).

Calibration Strategy	Data Source	Downstream CPC	CPC Gap vs. Oracle
Ground Truth Decay	Ground truth data	0.7411	Reference
20 equal width bins	TLD from GT data	0.7403	-0.0008
3 unequal width bins	TLD from GT data	0.7348	-0.0063
3 unequal width bins	Meta Movement Dist. Maps	0.7080	-0.0331

Table 5 Factorial Ablation and Shapley Value Contributions (Average across $N = 25$ held-out target cities).

Model Component	Ablation CPC Drop (%)	Shapley Value (CPC Gain)	Contribution (%)
Distance-Decay Behavior $f(d)$	-34.9%	0.2258	85.8%
Structural Attraction A_j	-4.7%	0.0208	7.9%
Origin Production O_i	-3.6%	0.0167	6.4%

behavior dominates the interaction process, recovering it from aggregate data is the key to zero-shot mobility prediction.

(Note: For completeness, the transferability of the origin production component O_i and the corresponding feature selection via SHAP stability analysis are detailed in Appendix B and Appendix C.)

5.2.3 Interpretation

Instead of learning complete OD matrices as monolithic mappings, the proposed framework explicitly isolates the behavioral mechanism governing destination choice.

The results suggest that the majority of predictive information required for gravity-based mobility prediction resides in the distance-decay component, whereas production and attraction primarily refine an already well-defined behavioral structure.

This observation provides an explanation for why aggregate trip-length distributions—despite containing no origin–destination identities—remain sufficient for recovering useful mobility information.

In other words, the success of the proposed survey-free calibration pipeline arises not because aggregate data reconstruct complete OD matrices, but because they accurately recover the dominant behavioral mechanism underlying those matrices.

5.3 RQ3: Can recovered behavior enable transferable gravity calibration?

To address the final research question, we evaluate whether our proposed aggregate-calibrated gravity model can serve as a deployable, survey-free alternative to traditional locally-calibrated frameworks. We compare our model against zero-shot neural architectures and locally calibrated structures across the 25 held-out target cities. The results of these comparative evaluations are summarized in Table 6.

We deliberately exclude graph-based neural transfer models (e.g., MPGCN, GODDAG, TransGM) from this comparative evaluation, as these models inherently require either a subset of target-city OD observations for fine-tuning or full target-city mobility graphs, rendering them incompatible with the strict zero-shot, survey-free deployment setting defined in our problem formulation. Instead, DeepGravity is included as the primary neural baseline, as its design allows for pure feature-based transfer without requiring target-city OD data. Furthermore, while alternative error metrics (e.g., RMSE, MAE, KL/JS divergence) provide complementary perspectives, we report CPC as the primary evaluation metric due to its established theoretical interpretation in mobility flow reconstruction (Lenormand et al. 2016).

Our most significant empirical result is that the proposed aggregate-calibrated model achieves statistical equivalence with the locally-calibrated

Fig. 4 Shapley value contributions of individual gravity model components to overall predictive accuracy (CPC). The distance-decay mechanism is overwhelmingly the dominant component, accounting for 85.8% of the total predictive gain within the investigated framework. This dominant role explains why highly compressed aggregate mobility summaries are sufficient to calibrate competitive predictive models.

Table 6 Multi-Baseline Comparison (25 Held-Out Target Cities)

Model	Mean CPC	Std Dev	Type	Features
Group 1: Survey-Free Deployment (No target OD data)				
Proposed Model (Survey-Free, Meta MDM)	0.6860	0.0391	Decomposed	6
Proposed Model (Survey-Free, National Fallback)	0.6896	0.0384	Decomposed	6
DeepGravity (Zero-Shot)	0.7529	0.0285	Neural E2E	27
Group 2: Oracle Diagnostic (True outflows O_i^{true})				
Proposed Model (Oracle O_i, Meta MDM)	0.7337	0.0256	Decomposed	6
Proposed Model (Oracle O_i, National Prior)	0.7398	0.0248	Decomposed	6
DeepGravity (Oracle O_i Zero-Shot)	0.7526	0.0283	Neural E2E	27
Group 3: Locally-Calibrated Baselines (Fitted on target cities)				
Naive Population Baseline	0.6758	0.0384	Baseline	0
Traditional Tanner Gravity	0.7382	0.0322	Structure-only	0
Traditional Exponential Gravity	0.6700	0.0350	Structure-only	0
Radiation Model	0.4851	0.0516	Param-free	0

Traditional Tanner baseline. Specifically, under oracle outflows, a Two One-Sided Test (TOST) of equivalence confirms that our model using Meta MDM decay (CPC 0.7337) is equivalent to the locally-fitted baseline (CPC 0.7382) within a tight margin of $\Delta_{\mathrm{eq}} = 0.01$ CPC ($p = 0.00326$), despite utilizing only aggregate telemetry. A Wilcoxon signed-rank test shows that although the minor performance difference is statistically significant (raw $p = 0.00737$), the gap is practically negligible. Additionally, the proposed model statistically outperforms the parameter-free Radiation Model baseline (Wilcoxon test with Bonferroni adjustment, $p < 10^{-6}$).

While the DeepGravity baseline achieves a marginally higher CPC (0.7529 vs. 0.6860), this comparison must be contextualized by their fundamentally different data requirements. DeepGravity operates as a supervised black-box model that requires historical OD matrices for training, which are unavailable in our target data-scarce environments. In contrast, PIGF achieves highly competitive performance under strict zero-shot, survey-free conditions using only privacy-preserving aggregate telemetry. Furthermore, PIGF explicitly recovers an interpretable behavioral mechanism, offering a level of transparency that end-to-end neural architectures cannot provide.

Beyond this primary equivalence result, the multi-baseline comparison yields several supporting insights. First, we observe that predicting true origin outflows (O_i) does not significantly alter the predictive accuracy of the DeepGravity model (CPC 0.7526 vs. 0.7529), as its neural architecture jointly optimizes production and destination selection. In contrast, the proposed gravity model demonstrates a strong dependency on production accuracy, where the introduction of oracle outflows narrows the performance gap, nearly matching the performance of the locally-fitted baseline.

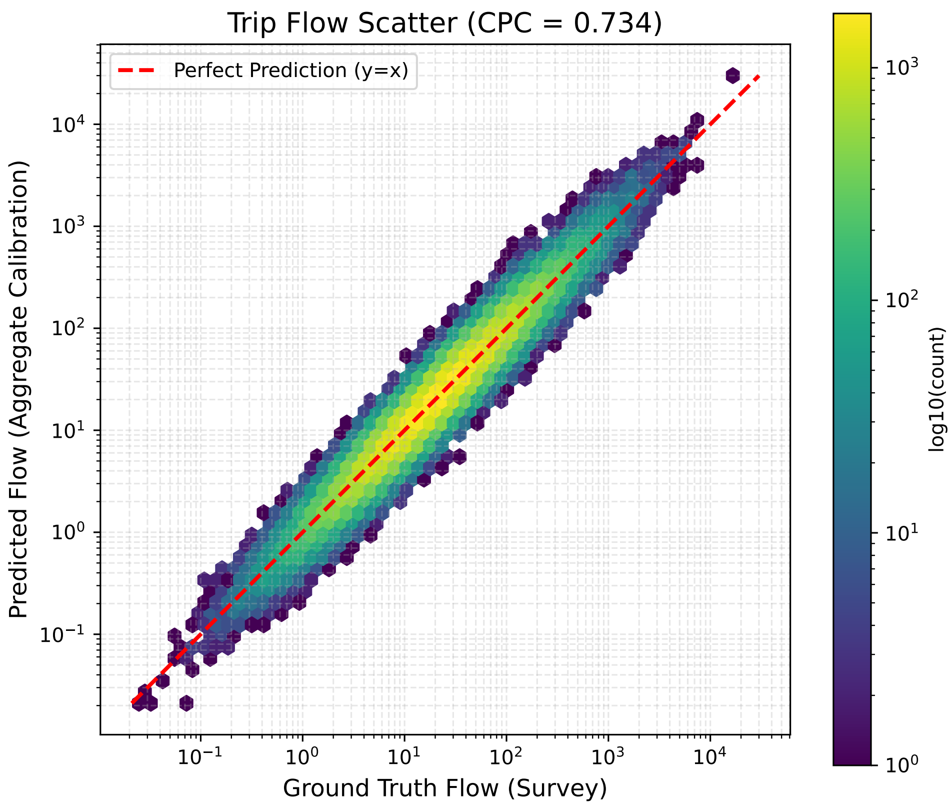

Fig. 5 Comparison of predicted flows versus ground-truth observations across the 25 target cities. The aggregate-calibrated model produces flow predictions that are statistically equivalent (TOST margin $\Delta_{eq} = 0.01$ CPC) to those produced by models calibrated directly on local origin–destination surveys.

5.3.1 Practical Implications

The proposed framework is intended for deployment in cities where conventional travel surveys are unavailable or prohibitively expensive.

Unlike existing gravity calibration approaches, PIGF requires only publicly available spatial datasets together with aggregate trip-length summaries that preserve user privacy.

Consequently, the framework provides a practical pathway for mobility estimation in data-scarce regions, privacy-sensitive environments, and rapidly developing cities where traditional OD surveys cannot be conducted routinely.

5.4 Methodological Validation

Beyond evaluating overall predictive performance, we conduct an exploratory analysis to examine whether urban morphology and density may influence the effectiveness of aggregate-based distance-decay calibration. Given the limited number of target cities and the single-country scope of the dataset, these analyses are intended to generate hypotheses for future investigation rather than establish causal relationships.

5.4.1 Impact of Urban Morphology

Preliminary patterns suggest that the physics-constrained gravity formulation achieves its highest predictive performance in Coastal environments (mean CPC of 0.714). The relative performance gap between the proposed gravity model and the neural DeepGravity baseline narrows to 5.8% in Coastal cities, compared to a larger 10.3% gap in Dense / Transit environments. The observations are consistent with physical geographical barriers restricting alternative routes and a greater explanatory role of distance decay in commuting flows. We note, however, that these morphology-specific findings should be interpreted cautiously due to several key statistical limitations: the overall sample size of target cities is

Table 7 Survey-Free Performance by Urban Morphology (25 Held-Out Target Cities)

Morphology Type	# Cities	Proposed CPC	DeepGravity CPC	Rel. Gap (%)
Dense / Transit	5	0.651	0.726	10.3%
Sprawling / Grid	10	0.691	0.760	9.0%
Polycentric	7	0.704	0.759	7.2%
Coastal	3	0.714	0.758	5.8%

small ($N = 25$), the morphology groups are highly unbalanced (e.g., only 3 Coastal cities and 5 Dense / Transit cities), and the resulting statistical power is limited.

5.4.2 Impact of Urban Density

To supplement this categorical classification, we analyze the continuous relationship between urban density and the performance gap ($\Delta\text{CPC} = \text{CPC}_{\text{DeepGravity}} - \text{CPC}_{\text{Proposed}}$). The gap correlates positively with both median employment density ($r = 0.263$) and population density ($r = 0.252$). The results are consistent with the hypothesis that dense, transit-oriented development decouples urban trips from simple Euclidean distance, allowing deep learning models to capture complex non-linear feature interactions, whereas lower-density or physically constrained environments follow more predictable distance friction behavior.

For instance, the metropolitan area of Atlanta serves as a prominent outlier in our evaluation, where its sprawling polycentric layout interacts poorly with the coarse 3-bin telemetry, causing a localized performance drop (detailed in Appendix A). Excluding such extreme outliers further improves the consistency of the behavioral recovery. In summary, these findings should be interpreted cautiously, but they suggest that our aggregate-calibrated model provides a robust alternative to locally-fit structures in low-density or physically constrained urban environments.

5.5 Scientific Findings

The experiments presented in this section were designed to validate the central scientific hypothesis introduced in Section 1. Rather than assessing only the final prediction accuracy, the evaluation systematically tested the mechanistic links: from aggregate mobility, to behavioral recovery, to gravity calibration, and finally to OD generation. Based on this progressive validation, we synthesize three core scientific findings:

1. **Finding 1 (Recoverability):** Aggregate mobility data preserve sufficient information to reliably recover latent city-specific distance-decay parameters without requiring any origin–destination observations.
2. **Finding 2 (Behavioral Dominance):** The recovered distance-decay behavior constitutes the dominant predictive mechanism governing spatial interaction, vastly outweighing the structural contributions of local production and attraction.
3. **Finding 3 (Practical Utility):** Zero-shot gravity calibration built upon this recovered behavior enables competitive survey-free OD estimation in unseen cities.

Collectively, the experiments support the central hypothesis: aggregate mobility contains sufficient behavioral information to substitute for OD matrices in model calibration. The principal contribution of PIGF is demonstrating this mechanistic pipeline: *Aggregate Mobility* $\rightarrow$ *Behavior Recovered* $\rightarrow$ *Gravity Calibrated* $\rightarrow$ *OD Generated*. By proving that the latent distance-dependent mechanism governing spatial interaction can be recovered from aggregate summaries and used for accurate flow prediction, these results motivate a broader discussion on how survey-free mobility modeling should be approached, which we explore in the following section.

6 Discussion

6.1 Scientific Implications: Recoverable Behavioral Information in Aggregate Mobility

The central finding of this study is not simply that PIGF achieves competitive survey-free OD prediction, but that aggregate trip-length distributions preserve sufficient information to recover the latent behavioral mechanism governing spatial interaction. Across the experimental analyses, aggregate mobility summaries consistently enabled accurate recovery of city-specific Tanner distance-decay functions, identified distance decay as the dominant predictive component within the investigated gravity formulation, and supported downstream OD prediction without requiring target-city mobility observations.

Crucially, this result clarifies the transportation-scientific interpretation of the recovered distance-decay parameters (α, β). In spatial interaction modeling, deterrence parameters should not be viewed as static, intrinsic traits of individual residents in isolation. Rather, they represent an emergent property of **human–environment interaction**:

$$\text{Transit \& Built Environment} \longrightarrow \text{Spatial Accessibility} \longrightarrow \text{Observed Travel Behavior } (\alpha, \beta) \tag{18}$$

Urban infrastructure, transit network density, and opportunity distributions shape regional spatial accessibility, which conditions how travelers navigate the city. By recovering (α, β) directly from aggregate trip-length distributions, the proposed framework captures a behavioral response consistent with the city's spatial structure. Taken together, these findings suggest that behavioral information can be inferred from aggregate mobility summaries to a much greater extent than has previously been assumed.

6.2 Extreme Information Compression and Behavioral Identifiability

A central theoretical insight emerging from our results is that recovering city-specific Tanner distance-decay parameters (α, β) does not require millions of disaggregate origin–destination (OD) flow observations. Instead, coarse aggregate trip-length distributions (TLDs)—even when compressed into as few as 3 physical distance bins (e.g., Meta MDM)—preserve sufficient information to reconstruct the underlying spatial interaction mechanism with near-lossless fidelity.

Empirically, the experimental results suggest that aggregate trip-length histograms retain sufficient information for reliable parameter identification. A primary factor is the parsimonious two-parameter formulation of the Tanner deterrence function together with exposure-corrected distribution matching. When disaggregate trips are aggregated into a K-bin probability vector $\mathbf{b} = (b_1, \ldots, b_K)^\top$ $(\sum b_k = 1.0)$, the histogram retains $K - 1$ independent degrees of freedom. For a 3-bin histogram ($K = 3$), the resulting system provides $K - 1 = 2$ independent empirical constraints, matching the two unknown parameters (α, β) in a well-posed system of non-linear equations.

Furthermore, the physical boundaries of aggregate telemetry products such as Meta MDM (0–10 km, 10–100 km, $>$ 100 km) provide optimal spectral isolation across travel scales. Short-distance trip fractions (0–10 km) are dominated by the power-law concentration parameter α, long-distance fractions ($>$ 100 km) are governed by exponential attenuation β, and the intermediate bin (10–100 km) anchors the structural transition. Consequently, aggregate telemetry acts as an information-preserving quadrature grid, explaining why survey-free 3-bin recovery achieves statistical equivalence with full ground-truth survey calibration ($\Delta_{\text{eq}} = 0.01$ CPC margin). This insight transforms mobility modeling by demonstrating that privacy-preserving, highly compressed mobility indicators are not coarse approximations, but information-preserving aggregate representations for behavioral inference.

6.2.1 From Flow Reconstruction to Behavioral Inference

Traditionally, survey-free mobility prediction has been approached as a flow reconstruction problem. Existing methods—including gravity calibration, deep learning, graph neural networks, and transfer-learning frameworks—primarily attempt to estimate complete origin–destination matrices, either by calibrating gravity parameters from observed OD flows or by learning direct mappings from historical mobility data. As a result, the availability of mobility observations remains central to most existing paradigms.

The findings of this study suggest an alternative perspective. Rather than reconstructing complete OD matrices directly, survey-free mobility

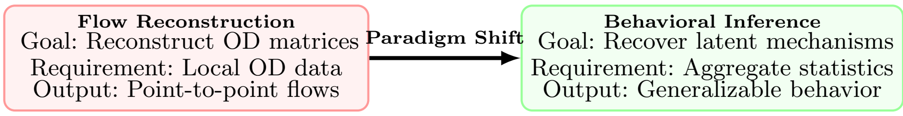

Fig. 6 The paradigm shift from direct flow reconstruction to behavioral inference. Rather than attempting to learn monolithic flow mappings from observed OD matrices, survey-free mobility modeling can alternatively recover the latent distance-dependent behavioral mechanisms underlying spatial interaction directly from aggregate summaries.

prediction may instead be formulated as a behavioral inference problem. Aggregate trip-length distributions do not retain origin–destination identities, yet they preserve the statistical signature of travel-distance preferences. Recovering this latent behavioral mechanism, and combining it with independently estimated production and attraction components, provides sufficient information to reconstruct meaningful flow patterns. This interpretation shifts the emphasis from recovering individual trips to recovering the behavioral process that generates those trips.

6.2.2 Why Aggregate Mobility Remains Informative

An important implication of the experimental findings is that the usefulness of aggregate mobility data does not arise from preserving detailed spatial trajectories, but from preserving the dominant behavioral constraints governing destination choice.

The Shapley analysis shows that distance decay accounts for the majority of predictive performance within the investigated gravity framework, substantially exceeding the contributions of production and attraction estimation. This provides a mechanistic explanation for why highly compressed trip-length distributions remain informative despite discarding nearly all spatial identities. In other words, aggregate mobility summaries appear to retain the behavioral information that matters most for gravity-based interaction, even when individual travel records are no longer available. This observation also helps explain why coarse aggregate products, such as three-bin movement distributions, continue to support effective calibration despite their limited spatial resolution.

6.2.3 Implications for Physics-Informed Mobility Modeling

The proposed framework also contributes to the broader development of physics-informed mobility modeling. Recent years have seen increasing interest in combining physical principles with machine learning to improve interpretability and transferability. Most existing approaches, however, incorporate physical constraints after mobility observations have already been collected for training. In contrast, PIGF uses the physical structure of the gravity model to infer behavioral parameters directly from aggregate mobility statistics, without requiring target-city OD observations.

This distinction suggests that physical models can serve not only as predictive constraints but also as identification mechanisms, allowing latent behavioral parameters to be recovered from privacy-preserving aggregate data. More broadly, these results illustrate how interpretable physical formulations and aggregate mobility products can complement, rather than compete with, modern data-driven approaches.

6.2.4 Scientific Perspective

Taken together, the experimental evidence supports a broader conceptual interpretation of survey-free mobility prediction. Rather than viewing aggregate mobility summaries as incomplete substitutes for origin–destination matrices, they may instead be regarded as observations of the underlying behavioral processes that govern spatial interaction. Under this perspective, the central task is no longer to reconstruct every individual movement, but to identify the dominant behavioral mechanism that shapes those movements.

Traditional mobility modeling asks: *"Can we reconstruct flows?"* This work asks a fundamentally different question: *"Can we recover the behavioral mechanism that generated those flows?"* If the underlying behavioral mechanism can be identified, origin–destination flow reconstruction becomes an emergent downstream consequence rather than the primary modeling objective.

6.3 Methodological and Practical Implications

Beyond its scientific implications, the proposed framework also has broader methodological and practical significance for urban mobility modeling. The experimental results suggest that the proposed decomposition strategy changes not only how distance-decay parameters are estimated, but also how survey-free mobility prediction can be constructed under severe data constraints.

6.3.1 Decomposition as a Modeling Strategy

Most existing OD prediction methods optimize a single model to approximate complete origin–destination matrices. Whether based on classical calibration, deep learning, or transfer learning, these approaches generally learn a monolithic mapping between spatial inputs and mobility flows.

The proposed framework follows a different strategy by explicitly decomposing spatial interaction into production, attraction, and behavioral distance decay, allowing each component to be estimated from the information source most closely associated with its physical meaning. This decomposition offers two methodological advantages. First, it enables each component to be estimated using the most appropriate information source rather than forcing all information into a single predictive model. Second, it produces a transparent inference pipeline in which each component can be independently validated, interpreted, and improved without retraining the entire system.

6.3.2 Aggregate Data as Calibration Signals

The results also suggest a broader role for aggregate mobility products. Current aggregate datasets are frequently viewed as reduced substitutes for detailed trajectory records because they discard origin–destination identities and individual travel histories. Under this perspective, aggregate data are often considered insufficient for model calibration.

The present findings suggest a different interpretation. Aggregate mobility summaries may instead be regarded as behavioral calibration signals rather than incomplete mobility observations. Although they do not reveal where individual trips begin or end, they preserve information about the collective travel behavior governing destination choice. Consequently, aggregate mobility products can support model calibration even when they are insufficient for direct flow reconstruction. This perspective may become increasingly relevant as privacy-preserving mobility products continue to replace individual-level trajectory datasets.

6.3.3 Implications for Survey-Free Mobility Prediction

From a practical perspective, the proposed framework substantially reduces the information required for gravity-model calibration. Instead of requiring household travel surveys, GPS trajectories, or mobile-phone records, calibration relies only on publicly available spatial datasets together with aggregate trip-length distributions. These data sources are considerably easier to obtain, more consistent across regions, and inherently more compatible with modern privacy requirements.

As a result, the framework provides a feasible pathway for estimating commuting flows in metropolitan areas where conventional mobility surveys are unavailable, prohibitively expensive, or restricted by legal or privacy considerations.

6.3.4 Scope of Applicability

The proposed framework is particularly suited to deployment scenarios characterized by limited mobility observations but reasonable availability of open spatial data. Examples include rapidly growing metropolitan regions, cities with infrequent household travel surveys, and privacy-sensitive applications where individual trajectories cannot be collected or shared.

Conversely, in data-rich environments where large-scale OD observations are routinely available, high-capacity supervised neural models may still achieve higher prediction accuracy. Under such circumstances, the primary advantage of PIGF lies in its interpretability, lower data requirements, and straightforward deployment rather than maximizing predictive performance.

Taken together, these methodological and practical implications suggest that the principal value of PIGF extends beyond the proposed implementation itself. By combining interpretable model decomposition with privacy-preserving aggregate mobility information, the framework illustrates a practical pathway toward survey-free urban mobility modeling that balances predictive performance, interpretability, and realistic data availability.

6.4 Scope and Limitations

Although the experimental results consistently support the proposed behavioral inference framework, the conclusions of this study should be interpreted within the scope of the present experimental setting and modeling assumptions. Recognizing these boundaries is important for understanding both the applicability of the proposed framework and the opportunities for future research.

6.4.1 Scope of the Experimental Validation

The empirical evaluation is conducted using 50 metropolitan areas in the United States, including 25 source cities for transferable production learning and 25 held-out target cities for evaluation. These metropolitan areas encompass diverse urban morphologies, ranging from dense transit-oriented cities to sprawling, polycentric, and coastal environments. Nevertheless, they remain embedded within a common national context characterized by similar census definitions, transportation systems, and data collection procedures.

Accordingly, the present experiments demonstrate robustness across heterogeneous U.S. metropolitan environments, but do not establish that the same behavioral relationships necessarily hold under substantially different institutional, socioeconomic, or transportation contexts. Additional evaluation across countries and mobility datasets will therefore be necessary before broader conclusions can be drawn.

6.4.2 Dependence on the Gravity Formulation

The proposed framework is built upon the classical gravity formulation, in which spatial interaction is decomposed into origin production, destination attraction, and distance-dependent behavioral friction. Consequently, the interpretation of recovered behavioral information is specific to this modeling framework.

Although the experiments demonstrate that distance decay is the dominant predictive component within the investigated gravity formulation, this conclusion should not be interpreted as evidence that all urban mobility systems are governed primarily by geographical distance. In metropolitan areas where multimodal accessibility, public transportation networks, or socioeconomic interactions play a stronger role, additional behavioral mechanisms may become equally or more important. This limitation reflects the scope of the current behavioral model rather than the behavioral inference framework itself.

6.4.3 Aggregate Mobility Representation

The present study evaluates behavioral recovery using aggregate trip-length distributions represented as one-dimensional histograms. Although such representations substantially reduce privacy exposure while preserving dominant travel-distance characteristics, they inevitably discard other dimensions of human mobility, including travel direction, temporal dynamics, transport mode, and activity purpose.

Consequently, the proposed framework should be interpreted as recovering the dominant distance-dependent component of aggregate mobility rather than the complete behavioral complexity of urban travel.

6.4.4 Survey-Free Does Not Mean Data-Free

Finally, although PIGF eliminates the need for target-city origin–destination observations, it should not be interpreted as a completely data-free approach. The framework still relies on publicly available spatial information, including population distributions, built-environment characteristics, and aggregate mobility summaries, to estimate the individual components of the gravity model.

Accordingly, the contribution of this work lies in reducing the dependence on localized mobility observations, rather than eliminating data requirements altogether. The proposed framework therefore complements, rather than replaces, data-rich mobility modeling when detailed observations are available.

Taken together, these limitations define the current scope of the proposed behavioral inference framework rather than diminishing its principal findings. Within the evaluated setting, the experiments consistently demonstrate that aggregate trip-length distributions contain sufficient information to recover behaviorally meaningful distance-decay functions and support survey-free gravity calibration. Extending this framework beyond the present assumptions and evaluation domain remains an important direction for future research.

6.5 Future Research Directions

The limitations identified above suggest several promising directions for future research. Rather than representing shortcomings specific to PIGF alone, they reflect broader challenges in survey-free mobility modeling and behavioral inference from aggregate data.

6.5.1 From Behavioral Recovery to Environmental Generation

While this study demonstrates that latent distance-decay parameters can be recovered from aggregate mobility summaries, the next fundamental scientific frontier is moving from behavioral recovery to environmental generation: modeling how the physical urban structure and built environment generate and induce travel behavior in the first place (Urban Structure & Infrastructure $\rightarrow$ Behavioral Friction $\rightarrow$ Observed Mobility). Investigating how transit network topology, spatial density, and opportunity distributions causally shape the recovered decay parameters (α, β) opens a compelling new research question at the intersection of urban morphology, transportation science, and spatial interaction theory.

6.5.2 Richer Behavioral Impedance Representations

While the current framework models travel behavior through geographical distance, future research may extend the behavioral inference framework to incorporate multi-dimensional impedance measures, including travel time, transit accessibility, monetary cost, and temporal congestion. Recovering multiple behavioral dimensions simultaneously may provide a more comprehensive description of urban mobility while preserving the survey-free philosophy of the proposed framework.

6.5.3 Cross-Country and Cross-Cultural Validation

The present evaluation is conducted on U.S. metropolitan areas. An important next step is to evaluate whether the proposed behavioral inference framework remains applicable across global cities with substantially different socioeconomic conditions, public transport infrastructures, and cultural travel patterns, determining whether behavioral recoverability represents a universal property across diverse urban systems.

6.5.4 Toward Hybrid Physics-Informed Learning

Finally, the explicit behavioral recovery formulation of PIGF provides a natural theoretical bridge to neural representation learning architectures such as neuroGravity (Yang et al. 2026). While neuroGravity demonstrates that urban environments shape implicit neural representations of mobility networks (Urban Environment $\rightarrow$ Neural Representation $\rightarrow$ Mobility), PIGF recovers explicit, exposure-corrected behavioral friction parameters (Urban Environment $\rightarrow$ Spatial Accessibility $\rightarrow$ Behavioral Response (α, β) $\rightarrow$ Mobility). Combining neural representation learning of urban socioeconomic and built-environment structure with explicit behavioral recovery from aggregate telemetry offers a promising path toward unified, physics-informed foundation models for urban spatial interaction.

Taken together, these directions indicate that the proposed framework should be viewed not merely as an engineering solution, but as an initial demonstration of behavioral inference from aggregate mobility summaries. Future developments may extend this paradigm toward modeling environmental behavior generation, broader geographical settings, and hybrid physics-informed learning frameworks.

7 Conclusion

Urban mobility modeling has traditionally depended on localized origin–destination (OD) observations for gravity-model calibration, limiting its applicability in data-scarce and privacy-sensitive environments. This study investigated a more fundamental question: whether aggregate mobility summaries preserve sufficient behavioral information to recover the latent distance-decay mechanism governing spatial interaction.

By reformulating survey-free mobility prediction as an inverse inference problem, this study demonstrated that the city-specific distance-decay parameters can be recovered directly from aggregate trip-length distributions via exposure-corrected information-theoretic distribution matching.

The experimental results across 50 US metropolitan areas validated three key findings. First, aggregate mobility summaries enable the accurate recovery of latent distance-decay behaviors. Second, game-theoretic analysis revealed that this recovered behavior constitutes the dominant predictive mechanism, contributing 85.8% of the predictive gain. Finally, integrating this behavioral component into a Projection–Inference Gravity Framework (PIGF) achieved zero-shot origin–destination flow predictions statistically equivalent to locally calibrated models.

These findings support a conceptual shift in mobility modeling: privacy-preserving aggregate mobility datasets are not merely incomplete substitutes for origin–destination matrices, but powerful behavioral calibration signals.

Ultimately, this study demonstrates that aggregate trip-length distributions preserve behaviorally meaningful responses that are consistent with city-specific spatial interaction patterns.

Appendix A Behavioral Recovery Evidence

A.1 Parameter Recovery Curves

Detailed visualizations of parameter convergence and cross-entropy optimization surfaces during the L-BFGS-B optimization.

A.2 Parameter Recovery across Morphologies

Statistical distribution of recovered α and β parameters stratified by the urban morphology classes defined in the main text.

A.3 Representative City Profiles

Case studies of behavioral recovery in specific metropolitan areas (e.g., Atlanta vs. Boston), highlighting how urban structure influences inference.

A.4 Behavioral Consistency Over Time

(Placeholder) Future longitudinal studies demonstrating the stability of the recovered distance-decay parameters across different temporal slices.

A.5 Distance-Bin Histogram Robustness

Analysis showing how varying the number of distance intervals K (e.g., 3, 5, 10, 20 bins) impacts the precision of behavioral inference.

A.6 Cross-City Distance Distribution Comparisons

Empirical density plots comparing observed aggregate trip-length distributions against the expected theoretical distributions generated by the recovered parameters.

Appendix B Methodological Validation

B.1 Full Factorial Ablation Results

Complete tabular results for all $2^3 = 8$ component coalitions evaluated during the Shapley value analysis.

B.2 Shapley Value Calculations

Mathematical details of the cooperative game formulation used to attribute predictive performance to individual gravity components.

B.3 Outflow Transferability (GBDT)

Performance metrics and learning curves for the zero-shot origin production model across different source-city training sizes.

B.4 Feature Selection Stability

SHAP summary plots justifying the reduction from the 37-candidate spatial feature set to the final 6 core variables.

B.5 Noise Sensitivity Analysis

Evaluation of downstream CPC degradation when additive Gaussian noise is injected into the intermediate production predictions.

B.6 Missing Attraction Data Robustness

Performance stability when varying percentages (10%, 20%, 50%) of destination attraction values are randomly masked.

B.7 CBD Collapse Simulation

Simulated resilience of the distance-decay recovery when the most attractive urban zones are subjected to artificial extreme density reductions.

Appendix C Reproducibility Details

C.1 Environment & Hardware Configuration

(Placeholder) Compute cluster specifications, GPU/CPU details, and software environments utilized for experiments.

C.2 Training Hyperparameters

Complete listing of GBDT hyperparameters, learning rates, tree depths, and early stopping criteria.

C.3 Random Seed Sensitivity

Variance analysis of model predictions across multiple random seed initializations.

C.4 Dataset Versioning

Specific timestamps, API versions, and access dates for OpenStreetMap, WorldPop, and Meta Movement Distribution Maps data.

C.5 Feature Engineering Pipeline

Step-by-step documentation of spatial aggregation, coordinate projection, and normalization procedures.

C.6 Code Repository Structure

(Placeholder) Overview of the accompanying open-source GitHub repository architecture.

C.7 Pre-trained Model Checkpoints

(Placeholder) Information regarding the availability and formats of pre-trained GBDT production models.

C.8 Open Data Access Restrictions

Details regarding licensing and geographical availability limitations for the utilized telemetry products.

C.9 Evaluation Metric Implementations

Mathematical formulas and boundary conditions used for CPC, RMSE, and MAE calculations within the evaluation codebase.

C.10 Troubleshooting Common Issues

(Placeholder) Solutions to potential numerical instabilities or data ingestion errors encountered during replication.

Appendix D Mathematical Foundations

D.1 MLE Derivation

Step-by-step derivation of the exposure-corrected log-likelihood objective function.

D.2 Parameter Identifiability Bounds

Theoretical conditions necessary for unique identifiability of the Tanner parameters from aggregate distributions.

D.3 Optimization Constraints

Log-space transformations used to strictly enforce positivity constraints ($\alpha \geq 0, \beta > 0$) during L-BFGS-B optimization.

D.4 Adaptive Geometric Offset Derivations

Proof illustrating how the adaptive offset δ maintains scale invariance across differing spatial zone resolutions.

D.5 Normalization Consistency Proofs

Demonstration that the multiplicative fusion step perfectly preserves origin-side trip production totals.

D.6 Gravity Factorization Details

Extended mathematical discussion on separating distance friction from opportunity distribution within spatial interaction matrices.

D.7 D7. Identifiability Discussion

A conceptual discussion bridging the methodological design and its theoretical underpinnings, detailing precisely why one-dimensional aggregate trip-length histograms preserve sufficient structural information to robustly identify the latent distance-decay parameters (β), even in the absence of origin–destination identities.

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